

Energy Shock and Firm-Level Decarbonisation

Did the Russia–Ukraine War Reduce GHG Intensity in European Industry?

Data Group 2

Marius Borsu Lefevre Giovanni Clemente Greta Guadalupi
Petr Hrebenar Stéphane Miquel Clément Pietro Montano
Souhail Habib Nehme Joan Olaf Subirachs Garcia

Bocconi University — Geopolitics for Business

May 2026

The context

Russia's 2022 invasion of Ukraine triggered the largest European energy supply disruption in decades, sending gas prices soaring and forcing firms to rethink their energy sourcing overnight.

- Europe had built production models around cheap Russian pipeline gas
- The shock arrived as climate commitments tightened: EU Fit for 55, CSRD
- Energy-intensive industries — chemicals, metals, cement, paper — were hit hardest
- Yet the literature on geopolitical shocks and environmental performance is thin

The gap

We know a lot about how geopolitical shocks affect investment and profitability. We know far less about their effect on **firm-level environmental performance**.

- Quantifying the firm-level GHG response is essential to evaluate whether energy-security shocks and climate policy *reinforce* or *undermine* each other
- Country-level energy infrastructure may become a *bottleneck* for decarbonisation — even when price incentives are strong
- Policy implication: energy-mix flexibility at the country level enables (or blocks) firm-level environmental adjustment

Did the Russia–Ukraine energy shock trigger a **reduction in firm-level GHG intensity** among European energy-intensive firms — and was this effect **different** for firms in countries more or less dependent on Russian gas?

- **Outcome:** GHG Direct Intensity — Scope 1 emissions (tCO₂e) per \$1M revenue
- Measured by **TruCost (S&P Global)**: captures *operational* emissions efficiency, normalised for firm size, with lowest measurement noise across datasets (NGFS 2024)

Panel dataset

- $\approx 973,000$ firm-year observations
- 28 European countries (EU-27 + NO, CH, UK)
- Years: 2018–2024 (pre-shock: 2018–2021; post-shock: 2022–2024)
- Publicly listed firms in manufacturing and resource sectors

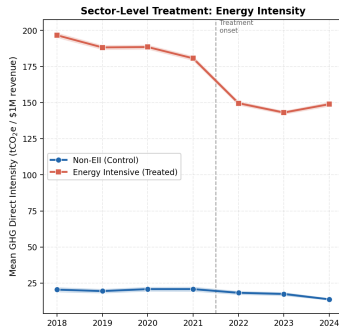
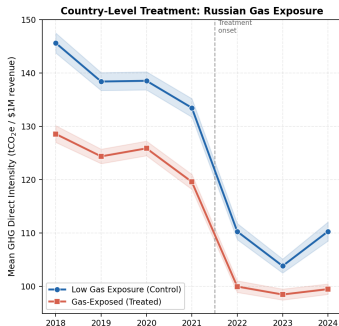
Sources

- **TruCost (S&P Global)** — GHG Direct Intensity (Scope 1)
- **World Bank** — GDP per capita, CPI inflation
- **Eurostat** — Russian gas import shares (`nrg_ti_gas`, `nrg_bal_c`)
- **EMAPS / Ember** — Country-year electricity prices

Identification Strategy: Difference-in-Differences

- **Core idea:** compare the change in GHG intensity for *treated* firms relative to *control* firms, before vs. after February 2022
- **Why the shock is exogenous:** the RU-UA war and energy disruptions were unanticipated by firms — no pre-announcement or anticipation bias
- **Parallel trends:** treated and control firms show broadly parallel GHG intensity trends in 2018–2021; validated via event-study pre-period coefficients

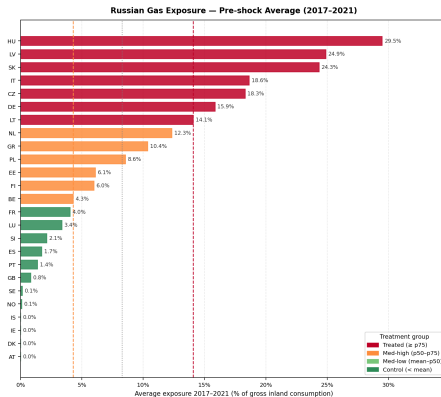
Parallel Trends — GHG Direct Intensity: Treated vs. Control
Sample: 2018–2024 | Shaded bands: $\pm 95\%$ CI | Raw means



Treatment 1: Country Gas Exposure

$$\text{Exp}_{c,t} = \frac{\text{RU gas imports}_{c,t}}{\text{Gross Inland Consumption}_{c,t}}$$

- Pre-shock average
2018–2021
- **Treated:** top quartile ($\geq$
p75 = 14.1%)
- 8 countries: HU, LV, SK,
IT, CZ, DE, LT, BG
- Source: Eurostat
`nrg_ti_gas`, `nrg_bal_c`



Avg. Russian gas exposure by country, 2018–2021. Red = treated ($\geq$ p75).

Model 1: Gas Exposure Specification

$$Y_{i,t} = \alpha_i + \lambda_t + \beta_1 \cdot \text{Post}_t \times \text{GasExp}_i + \gamma' X_{c,t} + \varepsilon_{i,t} \quad (1)$$

- $Y_{i,t}$: GHG Direct Intensity (tCO₂e / \$M revenue)
- α_i : firm fixed effects λ_t : year fixed effects
- $\text{GasExp}_i = 1$ if firm's country is in the top-p75 exposure group
- $\text{Post}_t = \mathbf{1}(t \geq 2022)$
- $X_{c,t}$: GDP per capita + CPI inflation (country-year controls)
- SE clustered at the **country level**

Coefficient of interest

$\hat{\beta}_1$: avg. change in GHG intensity for gas-exposed countries vs. non-exposed, post-2022 vs. pre-2022

Treatment 2: Energy-Intensive Sectors (EII)

Treated (EII = 1)

OECD/EC official EII classification:

- C16 Wood products
- C17 Paper
- C19 Coke & refined petroleum
- C20 Chemicals
- C22 Rubber & plastics
- C23 Non-metallic minerals (cement, glass)
- C24 Basic metals

Control (EII = 0)

Comparable manufacturing, lower energy input share:

- C21 Pharmaceuticals
- C25 Fabricated metals
- C28 Machinery
- C29 Motor vehicles
- C31 Furniture
- C18 Printing

66% of observations in EII sectors (647K firm-years)

Model 2: Sector Energy Intensity Specification

$$Y_{i,t} = \alpha_i + \lambda_t + \beta_2 \cdot \text{Post}_t \times \text{Ell}_i + \gamma' X_{c,t} + \varepsilon_{i,t} \quad (2)$$

- $\text{Ell}_i = 1$ for energy-intensive sectors (C16, C17, C19, C20, C22, C23, C24)
- Same firm/year FE and country-clustered SE as Model 1

Coefficient of interest

$\hat{\beta}_2$: avg. change in GHG intensity for EII firms vs. non-EII, post-2022 vs. pre-2022

Final Model: Triple Interaction (Model 3)

$$Y_{i,t} = \alpha_i + \lambda_t + \beta_1 \text{Post}_t \times \text{GasExp}_i + \beta_2 \text{Post}_t \times \text{Ell}_i + \beta_3 \text{Post}_t \times \text{GasExp}_i \times \text{Ell}_i + \gamma' X_{c,t} + \varepsilon_{i,t} \quad (3)$$

- $\hat{\beta}_1$: standalone country gas-exposure effect
- $\hat{\beta}_2$: standalone Ell sector effect
- $\hat{\beta}_3$: **additional** effect for firms that are *both* in gas-exposed countries *and* in Ell sectors

Why triple interaction?

Allows us to decompose country- and sector-level channels separately, and test whether the two dimensions of treatment reinforce or offset each other.

Results: All Models

	(1) Gas Exposed	(2) Energy Intensive	(3) Triple
Post $\times$ GasExp	5.209*** (1.089)	-38.461*** (2.209)	-2.611 (2.498)
Post $\times$ EI			-43.480*** (1.643)
Post $\times$ Both			9.863*** (2.543)
Observations	972,979	972,979	972,979
Within R^2	0.011	0.048	0.049
Firm / Year FE	Yes	Yes	Yes
SE clustered	Country	Country	Country

Dep. var.: GHG Direct Intensity (tCO₂e / \$1M revenue). SE in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

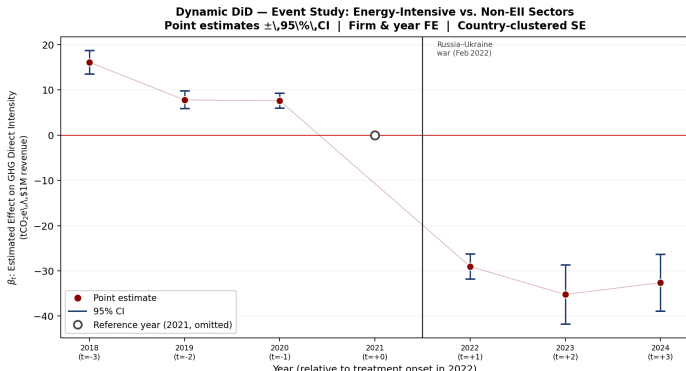
Finding: EII Firms Reduced GHG Intensity

Model 2

EII firms reduced GHG Direct Intensity by $\approx 38.5 \text{ tCO}_2\text{e}/\M post-2022 vs. non-EII firms

($p < 0.01$; $\approx 32\%$ of sample mean of $121 \text{ tCO}_2\text{e}/\M)

- Higher energy prices forced diversification away from gas $\rightarrow$ Scope 1 falls
- Effect is sustained across 2022–2024, not a one-off adjustment



Finding: Gas Dependency Attenuates Decarbonisation

Model 3

$\hat{\beta}_2 = -43.5^{***}$, $\hat{\beta}_3 = +9.8^{***} \Rightarrow$ net effect (doubly treated): $-33.7 \text{ tCO}_2\text{e}/\M

- Gas-dependent countries faced **constrained transition paths** (LNG bottlenecks, long renewable lead times)
- Standalone gas-exposure effect: $+5.2$ in Model 1 $\rightarrow$ indistinguishable from zero (-2.6 , $p = 0.30$) in Model 3

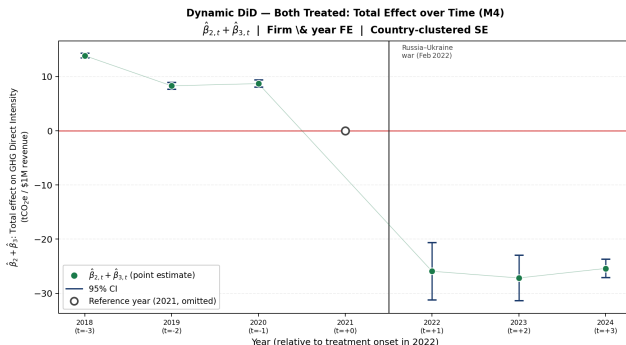


Figure: Dynamic DiD: total effect for doubly treated firms / $\hat{\beta}_2$ Ref: 2021

Discussion: Mechanism

- 1 Energy price shock → strong incentive to cut gas consumption
- 2 Firms diversify: energy efficiency, lower-carbon electricity, supply-chain changes
- 3 Gas is itself a GHG fuel → diversification *mechanically* reduces Scope 1

Country	Energy endowment	Adjustment capacity
France	Nuclear flexibility	Higher
Germany	Coal fallback + renewables	Moderate
Italy	LNG ramp-up constrained	Lower
HU / SK	High pipeline dependency	Lowest

Firm-level decarbonisation and **country-level energy infrastructure** are deeply complementary.

① HQ $\neq$ operations for MNEs

A German-registered firm producing in Poland faces the Polish energy mix. Facility-level data would be needed to correct this.

② Carbon leakage not identified

Firms may have outsourced emission-intensive steps abroad. Scope 3 or trade-adjusted accounting required.

③ Pre-trend caveat (EII treatment)

2018 pre-period coefficient elevated ($\approx +16$); results capture *acceleration* of a divergence, not its initiation. Post-shock drop (-25 to -35) is an order of magnitude larger.

④ Short post-shock window

Panel ends in 2024 — only two full post-shock years. Full trajectory may materialise as renewable capacity investments complete.

Main finding

The RU-UA energy shock **significantly reduced GHG intensity** among European EII firms: -38 – -43 tCO₂e/\$M (32–36% of sample mean). Effect is **smaller** in gas-dependent countries (≈ -10 tCO₂e/\$M attenuation).

Policy implications

Country level

Energy diversification (interconnectors, LNG, renewables) is both an energy-security *and* decarbonisation imperative.

Firm level

Prioritise energy-mix flexibility and long-term power-purchase agreements to decouple GHG performance from host-country endowments.